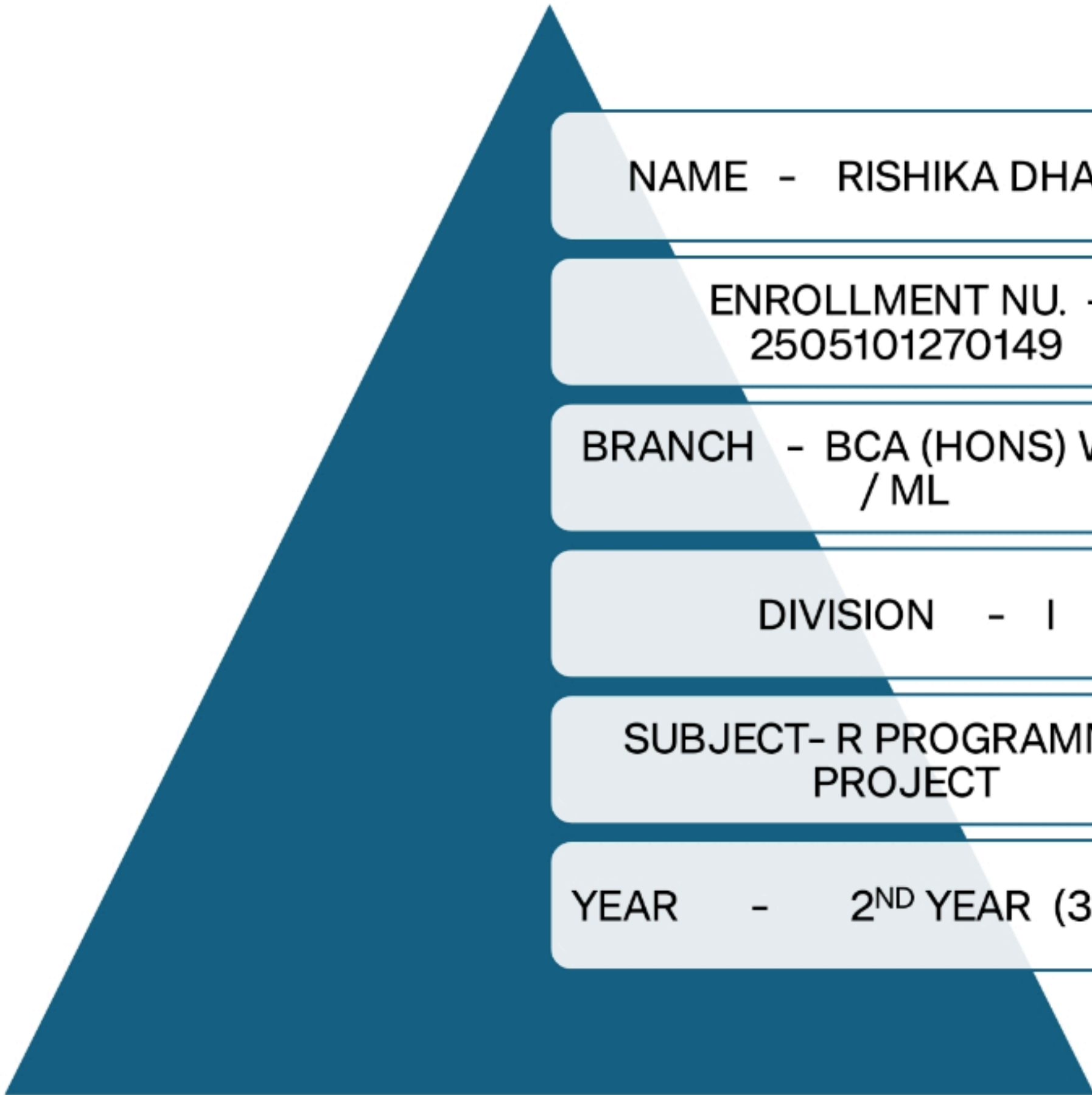




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BRANCH - BCA (HONS) WITH AI
/ ML

DIVISION - I

SUBJECT- R PROGRAMMING
PROJECT

YEAR - 2ND YEAR (3RD SEM)

 Source on Save

```
1 # Load built-in dataset
2 data(mtcars)
3
```

Type `q()`

```
> data(mtcars)
> head(mtcars)
```

	mpg	wt	qsec	vs	am	gear	carb
1	21.0	2.62	16.46	0	1	4	4

```
3
4 # View first few rows
5 head(mtcars)
6
```

```
> data(mtcars)
> head(mtcars)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

```
> str(mtcars)
```

```
data frame with 32 obs of 11 variables
```

Search

```
6
7 # Check structure
8 str(mtcars)
9
```

```
valiant      18.1      0      22.8 ...
> str(mtcars)
'data.frame':   32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num   6  6  4  6  8  6  8  4  4  6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110  93 110 175 105 245  62  95 123 ...
 $ drat: num   3.9  3.9  3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num   2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num   0  0  1  1  0  1  0  1  1  1 ...
 $ am  : num   1  1  1  0  0  0  0  0  0  0 ...
 $ gear: num   4  4  4  3  3  3  3  4  4  4 ...
 $ carb: num   4  4  1  1  2  1  4  2  2  4 ...
> summary(mtcars)
```

mpg

cyl

disp

hp

drat

Q Search

```

9
10 # Summary statistics
11 summary(mtcars)
12

```

39
26:1 (Top Level) 

Console Terminal Background Jobs

R - R 4.6.1 - ~/

```

$ carb: num 4 4 1 1 2 1 4 2 2 4 ...
> summary(mtcars)

```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear
Min. :10.40	Min. :4.000	Min. : 71.1	Min. : 52.0	Min. : 2.760	Min. :1.513	Min. :14.50	Min. :0.0000	Min. :0.0000	Min. :3.000
1st Qu.:15.43	1st Qu.:4.000	1st Qu.:120.8	1st Qu.: 96.5	1st Qu.:3.080	1st Qu.:2.581	1st Qu.:16.89	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:3.000
Median :19.20	Median :6.000	Median :196.3	Median :123.0	Median :3.695	Median :3.325	Median :17.71	Median :0.0000	Median :0.0000	Median :4.000
Mean :20.09	Mean :6.188	Mean :230.7	Mean :146.7	Mean :3.597	Mean :3.217	Mean :17.85	Mean :0.4375	Mean :0.4062	Mean :3.688
3rd Qu.:22.80	3rd Qu.:8.000	3rd Qu.:326.0	3rd Qu.:180.0	3rd Qu.:3.920	3rd Qu.:3.610	3rd Qu.:18.90	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:4.000
Max. :33.90	Max. :8.000	Max. :472.0	Max. :335.0	Max. :4.930	Max. :5.424	Max. :22.90	Max. :1.0000	Max. :1.0000	Max. :5.000

```

> sum(is.na(mtcars))

```

```
12  
13 # Check missing values  
14 sum(is.na(mtcars))  
15
```

```
Max. :0.000  
> sum(is.na(mtcars))  
[1] 0  
sum(mtcars$mpg, mtcars$hp)
```

```
15  
16 # Correlation between variables  
17 cor(mtcars$mpg, mtcars$hp)  
18
```

```
> sum  
[1] 0  
> cor(mtcars$mpg, mtcars$hp)  
[1] -0.7761684  
cor(mtcars$mpg)
```

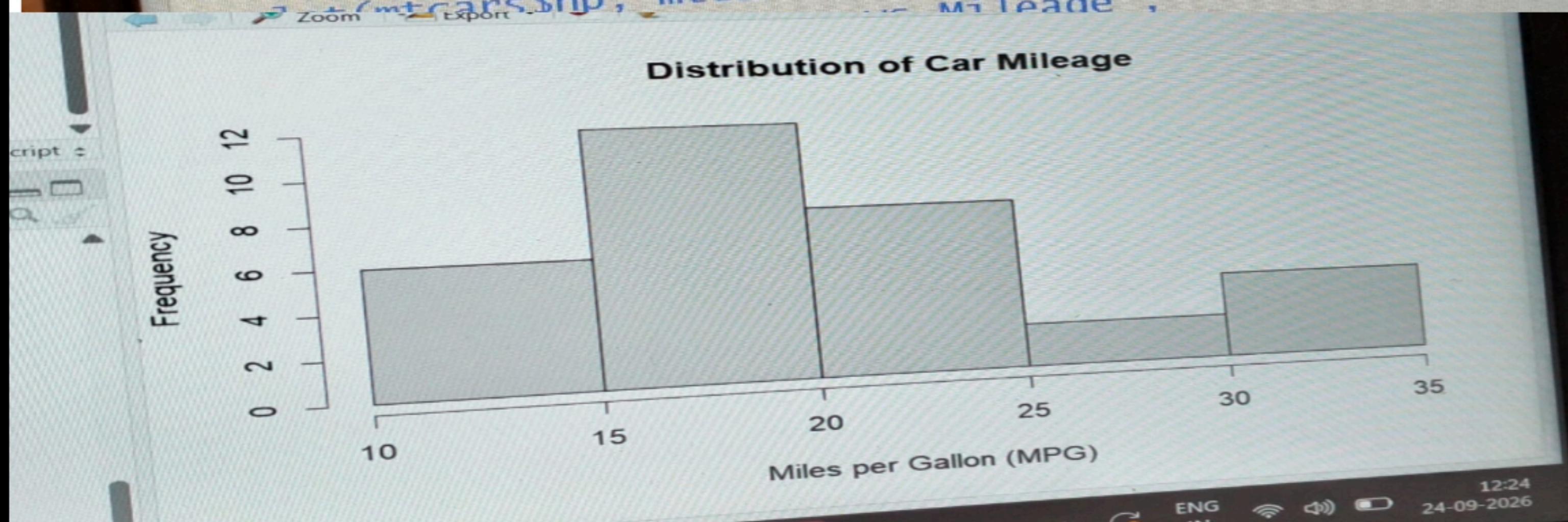
```
18  
19 # Average MPG  
20 mean(mtcars$mpg)  
21
```

```
[1] -0.7761684  
> mean(mtcars$mpg)  
[1] 20.09062  
> hist(mtcars$mpg,  
       # = "Distribution of Car Mileage",  
       # = (mpg))
```

```

21
22 # Histogram of MPG
23 hist(mtcars$mpg,
24       main = "Distribution of Car Mileage",
25       xlab = "Miles per Gallon (MPG)")
26 |
> mean(mtcars$mpg,
[1] 20.09062
> hist(mtcars$mpg,
+       main = "Distribution of Car Mileage",
+       xlab = "Miles per Gallon (MPG)")
+       mtcars$hp, mtcars$mpg,
+       main = "Mileage",

```




```
32  
33 # Bar chart of cylinders  
34 barplot(table(mtcars$cyl),  
35           main = "Number of Cars by Cylinders",  
36           xlab = "Cylinders",  
37           ylab = "Number of Cars")  
38
```

```
+ ylab = "MPG")  
+  
> barplot(table(mtcars$cyl),  
+           main = "Number of Cars by Cylinders",  
+           xlab = "Cylinders",  
+           ylab = "Number of Cars")  
+  
>
```

